

Curriculum Vitae: Juyeop Baek

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EDUCATION

UNIVERSITY OF TEXAS, AUSTIN

Austin, Texas

PhD in Electrical and Computer Engineering

August 2026 – May 2031 (Expected)

Advisor: Dr. David Z. Pan

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, Georgia

Master of Science in Electrical and Computer Engineering

August 2024 – May 2026

GPA: 4.0/4.0

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, Georgia

Bachelor of Science in Electrical Engineering

August 2018 – May 2022

GPA: 3.86/4.0

PROFESSIONAL RESEARCH EXPERIENCE

GRADUATE RESEARCH ASSISTANT, PSYLAB

June 2025 – May 2026

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, Georgia

- Advisor: Dr. Visvesh Sathé
- Thesis Title: Methodologies for Automated Power Delivery Network Extraction in Heterogeneous 2.5D/3D Packaging
- Automated PDN extraction flows for advanced packaging ASIC systems, optimizing granularity and power integrity analysis from VR and decoupling capacitor placement
- Constructed and examined end-to-end VR-to-load PDN models, comparing side-mounted, backside, and land-side architectures for efficiency

GRADUATE RESEARCH ASSISTANT, GTCAD

August 2024 – May 2025

GEORGIA INSTITUTE OF TECHNOLOGY

{Relocated to University of Southern California}

- Advisor: Dr. Sung Kyu Lim
- Created and verified AI/ML-assisted EDA flows for ASIC design in placement, routing, and S/P/T integrity testing

UNDERGRADUATE RESEARCH ASSISTANT, GAMMA LABORATORY

August 2020 – May 2022

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, Georgia

- Advisor: Dr. Shaolan Li
- Created an RTL-to-GDSII tool flow for a delta-sigma analog-digital converter made completely from EDA - friendly digital cells
- Leveraged Synopsys and Cadence software to generate and modify floorplans from Verilog descriptions while optimizing timing, area, and power

PUBLICATIONS

- [1] Jin Woong Kwak*, **Juyeop Baek***, Muhannad Bakir, Visvesh Sathe, “Modeling and Optimization of 2-stage Power Delivery Systems for High-Performance, Large-Area Packages”, *IEEE 76th Electronic Components and Technology Conference (ECTC)*, 2026 (***Equal Contribution**)
- [2] Aaron Jezghani, **Juyeop Baek**, Deepa Phanish, Michael D. Weiner, and Jeffrey Young. 2026. Sustainable On-Demand Computing for High-Performance Computing through Managed Oversubscription. In Proceedings of the 32nd International Conference on Job Scheduling Strategies for Parallel Processing (JSSPP’26).

TEACHING EXPERIENCE

Georgia Institute of Technology – Teaching Assistant (2020 – 2022)

- *Digital Design Laboratory*: Design and implementation of digital systems, including a FPGA-based hardware design project. CAD tools, project design methodologies, logic synthesis, and assembly language programming
- *Measurements, Circuits, and Microelectronics Laboratory*: Theory and experiments related to the design, analysis, construction, and measurement of elementary passive and active analog circuits using both discrete and integrated devices
- *Intro Physics II*: A calculus-based course with laboratory covering electromagnetism, applications of light and modern physics

PROFESSIONAL EXPERIENCE

PARTNERSHIP FOR AN ADVANCED COMPUTING ENVIRONMENT

August 2025 – Present

Georgia Institute of Technology

Atlanta, Georgia

- Assisted in the management and development of over 1000 GPU-enabled HPC datacenter
- Created various MIG and sharding scripts for an efficient workload partition and monitoring for efficient power consumption while allowing multiple concurrent processes

HARDWARE ENGINEERING INTERN, SAMSUNG ELECTRONICS

June 2024 – August 2024

Mobile eXperience Division, Display Group

Suwon, South Korea

- Specialized in the design of the display driver interface chip
- Created pass transistor and dynamic logic variations of circuits for higher propagation speed and lower power consumption

CONSULTANT INTERN, DELOITTE

March 2024 – June 2024

Monitor Deloitte, Consulting Division

Seoul, South Korea

- Conducted extensive research and analysis on the current state of the AI accelerator market, as well as the emerging new technologies
- Assisted in the development of client proposals and presentations by streamlining data collection processes

SKILLS

Programming

C++, CUDA, Python, Bash, Tcl, SKILL, Verilog, VHDL, MATLAB, PyTorch, Tensorflow

CAD	Cadence: Genus, Innovus, Virtuoso, Spectre, APD Synopsys: Design Compiler, IC Compiler II, HSPICE FPGA: Xilinx Vivado, Xilinx Vitis, Intel Quartus Siemens: Xpedition Enterprise, Hyperlynx, Calibre, Mentor Graphics Ansys: Electronics Desktop Suite, SiWave, Lumerical Keysight ADS
Hardware	FPGAs, Spectrum Analyzers, Oscilloscopes, Microcontrollers
Languages	English(Native), Korean(Native)

AWARDS & FUNDING

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| ▪ SRC Scholar (JUMP 2.0 CHIMES) | 2024 - Present |
| ▪ Faculty Honors | 2018 - 2022 |
| ▪ Dean's List | 2018 - 2022 |

PROFESSIONAL MEMBERSHIPS

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| ▪ Georgia Tech IEEE Chapter | 2018 – 2022 |
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LEADERSHIP AND SERVICE

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| ▪ Chair, Georgia Tech IEEE Chapter | August 2019 – May 2022 |
| ▪ Signal Intelligence Leader, Republic of Korea Army | September 2022 – March 2024 |